

CHAPTER 19

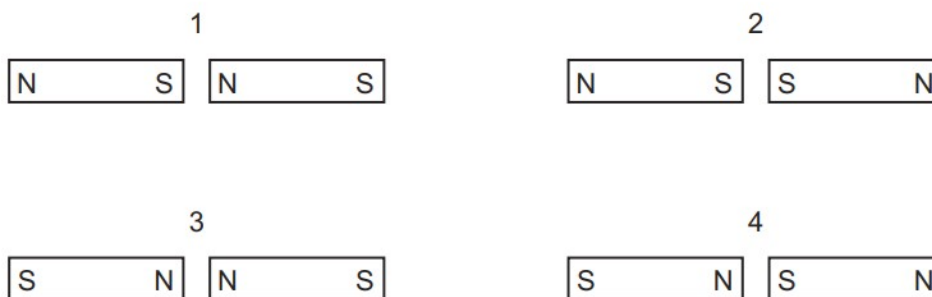
SIMPLE MAGNETISM

4.1 Simple magnetism and magnetic fields

- 1 Describe the forces between magnetic poles and between magnets and magnetic materials, including the use of the terms north pole (N pole), south pole (S pole), attraction and repulsion, magnetised and unmagnetised
- 2 Describe induced magnetism
- 3 State the difference between magnetic and non-magnetic materials
- 4 State the differences between the properties of temporary magnets (made of soft iron) and the properties of permanent magnets (made of steel)
- 5 Describe a magnetic field as a region in which a magnetic pole experiences a force
- 6 Describe the plotting of magnetic field lines with a compass or iron filings and the use of a compass to determine the direction of the magnetic field
- 7 Draw the pattern and direction of the magnetic field lines around a bar magnet
- 8 State that the direction of the magnetic field at a point is the direction of the force on the N pole of a magnet at that point
- 9 Know that the relative strength of a magnetic field is represented by the spacing of the magnetic field lines
- 10 Describe uses of permanent magnets and electromagnets

1. O/N/23/12/Q32

Two permanent magnets are placed side by side in four different arrangements.



In which arrangements do the magnets repel each other?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

2. M/J/23/12/Q24

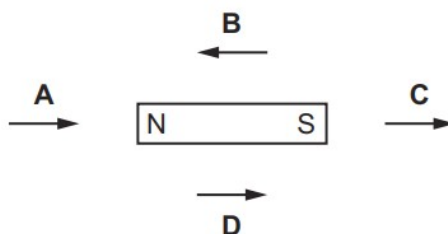
What proves that a metal bar is a permanent magnet?

- A** It attracts both ends of a compass needle.
- B** It attracts one end of another magnet.
- C** It conducts electricity.
- D** It repels one end of another magnet.

3. M/J/23/12/Q25

The diagram shows a bar magnet and four labelled arrows.

Which arrow shows the direction of the magnetic field at the point where the arrow is shown?



4. M/J/23/11/Q25

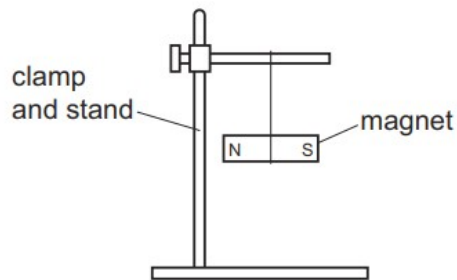
Bar magnets and various non-magnetic and demagnetised metal bars are placed in the different arrangements shown.

In which arrangement do the bars repel?



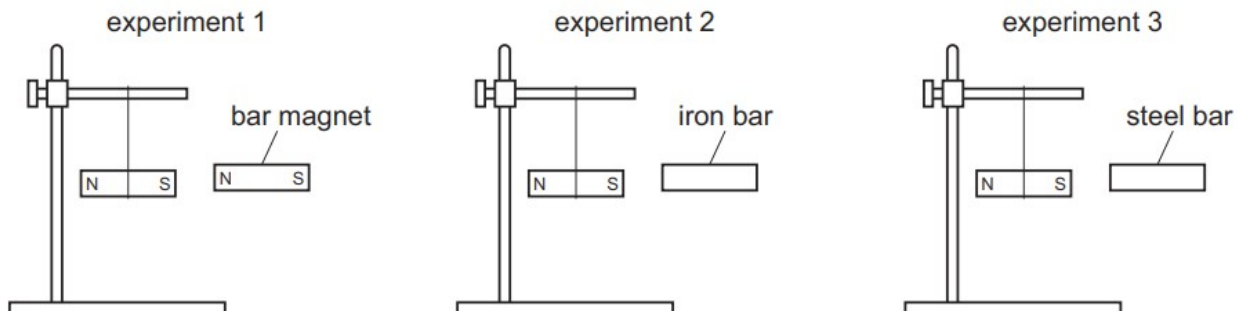
5. M/J/23/11/Q26

A student uses a wooden clamp and a stand to hold a freely suspended bar magnet.



The diagrams show that:

- in experiment 1, the student holds a bar magnet next to the suspended magnet
- in experiment 2, the student holds an iron bar next to the suspended magnet
- in experiment 3, the student holds a steel bar next to the suspended magnet.



In which experiments does the suspended magnet move?

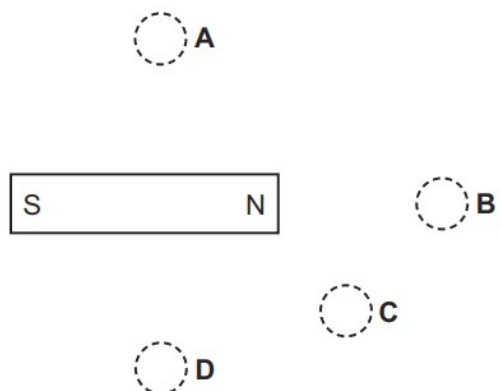
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

6. O/N/22/11/Q26

Four plotting compasses are placed near a bar magnet. Ignore any effects of the Earth's magnetic field.

One compass appears like this .

What is a possible position for this compass?



7. O/N/22/12/Q29

The N pole of a permanent magnet is placed in turn close to a piece of iron, the S pole of another magnet, and a piece of copper.

Which objects are attracted to the N pole of the permanent magnet?

	piece of iron	S pole	piece of copper
A	✓	✓	x
B	x	✓	x
C	x	✓	✓
D	x	x	✓

key

✓ = attracted

x = not attracted

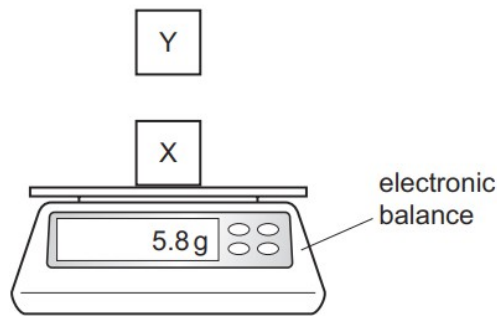
8. M/J/22/11/Q27

Which material is used for the core of an electromagnet?

- A** aluminium
- B** copper
- C** iron
- D** steel

9. O/N/21/11/Q27

Object X is on an electronic balance. Object Y is held above object X as shown.



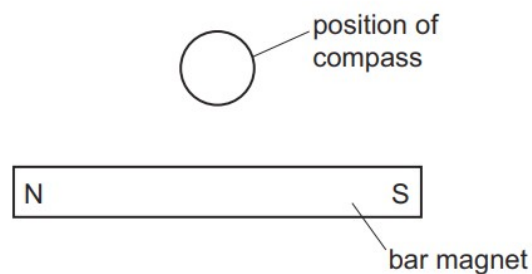
The reading on the balance increases when Y is moved closer to X.

Why does this happen?

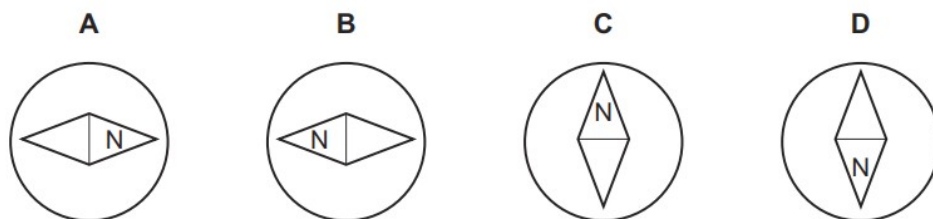
- A** X and Y are both made of iron.
- B** X and Y are both magnets.
- C** Y is a magnet and X is made of iron.
- D** Y is made of iron and X is a magnet.

10. O/N/21/12/Q30

A small plotting compass is placed near to a bar magnet as shown.



Which diagram shows the direction in which the compass needle points?



11. M/J/21/12/Q30

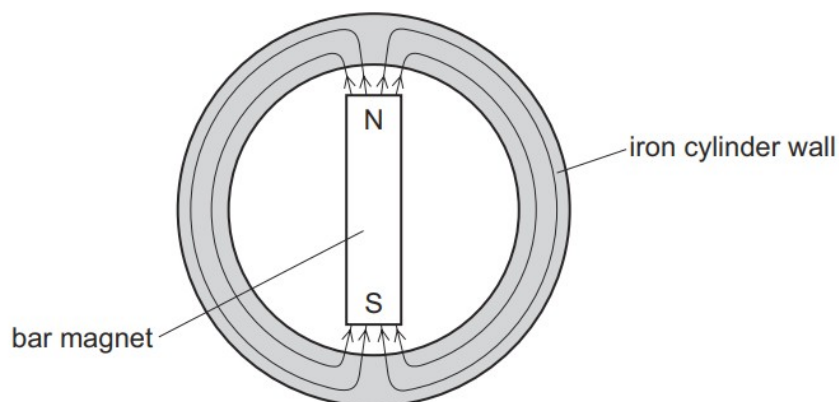
End X of a metal rod attracts the North pole of a compass needle.

Which statement about the rod is correct?

- A** It is made of copper that is not initially magnetised.
- B** It is made of copper with a South pole at X.
- C** It is made of steel that is not initially magnetised.
- D** It is made of steel with a North pole at X.

12. O/N/20/11/Q32

A bar magnet is placed in a hollow iron cylinder. The diagram shows the magnetic field pattern produced.

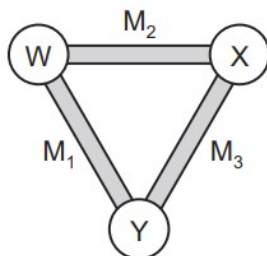


What is the cause of the field inside the cylinder wall?

- A** electromagnetic generation
- B** electromagnetic induction
- C** electrostatic induction
- D** magnetic induction

13. O/N/19/11/Q30

Magnets M_1 , M_2 and M_3 are joined by iron balls W, X and Y as shown.



Magnetic poles are induced in the iron balls W and X as shown.

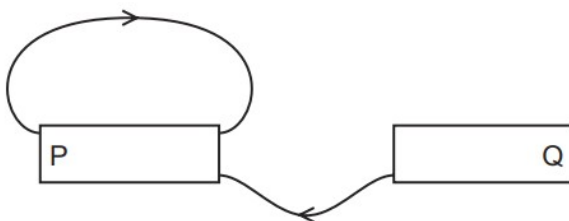


Which diagram shows the poles induced in ball Y?



14. O/N/19/12/Q29

The diagram shows two magnetic field lines of the magnetic field around two bar magnets.



Which row shows the magnetic poles at end P and end Q of the two magnets?

	P	Q
A	N	N
B	N	S
C	S	N
D	S	S

15. M/J/19/12/Q31

One end of each of two bars is marked with an X or a Y, as shown.



bar 1



bar 2

End X of bar 1 is brought close to the N-pole and then to the S-pole of a magnet.

End Y of bar 2 is brought close to the N-pole and then to the S-pole of the magnet.

The table shows what happens.

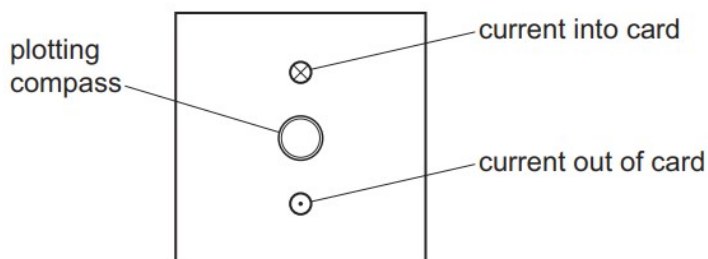
	end X of bar 1	end Y of bar 2
N-pole of magnet	attracted	attracted
S-pole of magnet	attracted	repelled

What materials are bar 1 and bar 2 made from?

	bar 1	bar 2
A	copper	iron
B	iron	steel
C	steel	copper
D	steel	iron

16. M/J/19/11/Q29

Two vertical wires pass at right-angles through a piece of card. There is a large current in each wire in the direction shown.



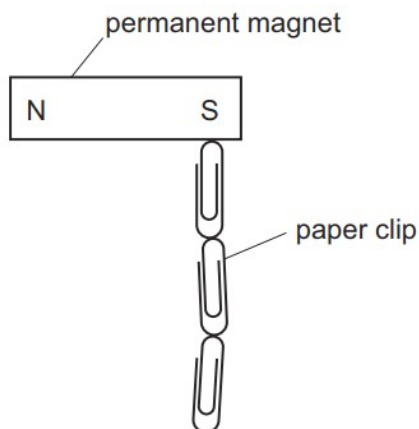
A plotting compass is placed on the card.

Which diagram shows the direction in which the needle of the plotting compass points?



17. M/J/19/11/Q28

A student investigates a permanent magnet by suspending paper clips from the magnet as shown.

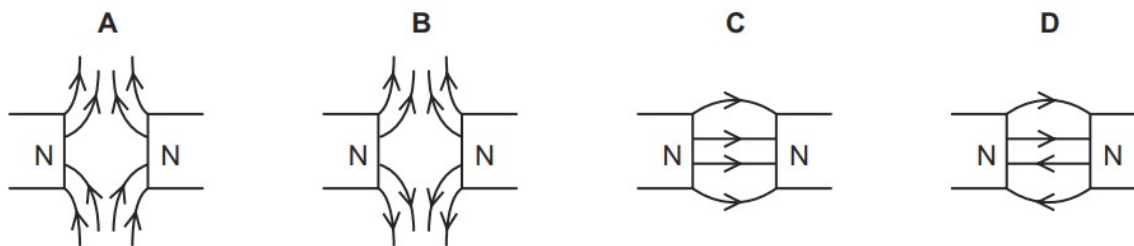


Which statement is correct?

- A** Each paper clip has a S-pole at the top and a N-pole at the bottom when suspended from the permanent magnet.
- B** If the paper clips are made of soft iron they become demagnetised when the permanent magnet is removed.
- C** If the paper clips are made of steel they become demagnetised when the permanent magnet is removed.
- D** More paper clips can be suspended from the N-pole.

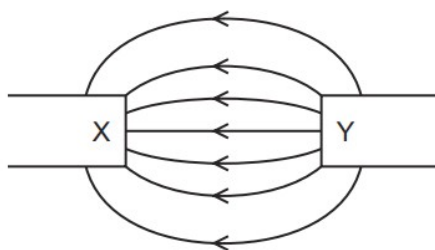
18. O/N/18/12/Q28

Which diagram shows the magnetic field pattern in the region between the N-poles of the two bar magnets?



19. O/N/17/11/Q26

The diagram shows the pattern and the direction of the magnetic field between two magnetic poles X and Y.



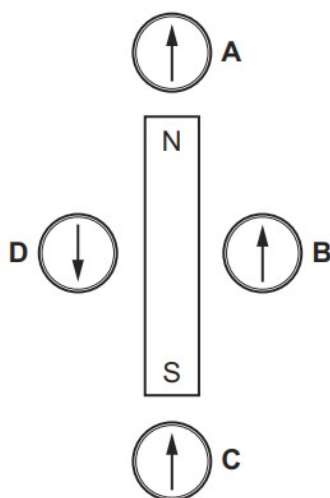
Which types of pole are X and Y?

	X	Y
A	N-pole	N-pole
B	N-pole	S-pole
C	S-pole	N-pole
D	S-pole	S-pole

20. O/N/17/12/Q26

A compass needle is moved to different places around a strong bar magnet.

Which position shows **incorrectly** the direction of the compass needle?



21. M/J/17/12/Q30

What proves that a metal bar is a permanent magnet?

- A** It attracts both ends of a compass needle.
- B** It attracts one end of another magnet.
- C** It conducts electricity.
- D** It repels one end of another magnet.

22. O/N/15/12/Q28

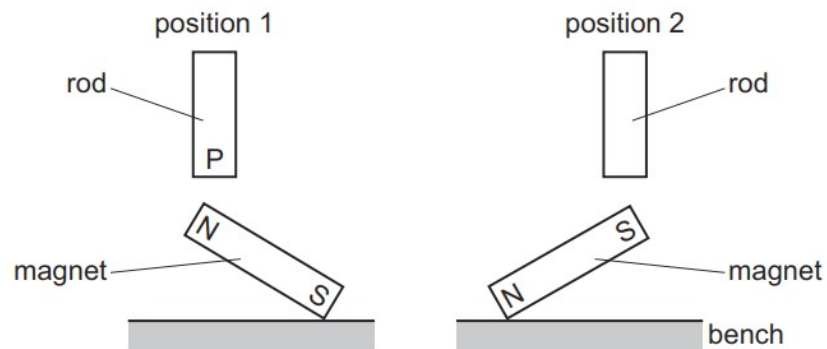
Which material can be picked up by a magnet?

- A** aluminium
- B** copper
- C** iron
- D** plastic

23. M/J/15/11/Q25

One end of a rod picks up the N-pole of a bar magnet when in position 1.

The same end of the rod picks up the S-pole of the bar magnet when in position 2.



Which material is the rod made from and what is the pole at end P of the rod when in position 1?

	material	pole at P
A	iron	N-pole
B	iron	S-pole
C	steel	N-pole
D	steel	S-pole

24. O/N/14/11/Q26

A metal bar PQ hangs from a thin thread and always comes to rest with end P pointing north. Another bar XY of the same metal settles in no definite direction.

What happens if the two bars are brought near one another?

- A End P and end Q both attract end X.
- B End P attracts end X but repels end Y.
- C End P neither attracts nor repels end X.
- D End P repels end X but attracts end Y.

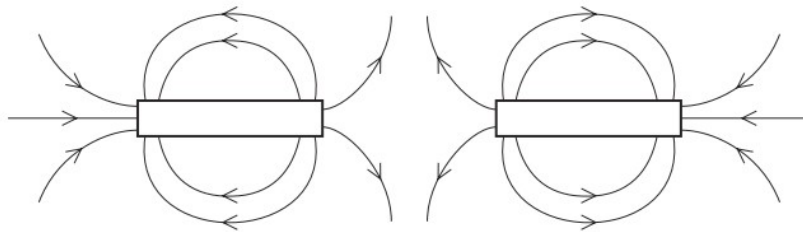
25. M/J/14/12/Q30

What always produces a permanent bar magnet?

- A an iron bar in a coil carrying alternating current (a.c.)
- B an iron bar in a coil carrying direct current (d.c.)
- C a steel bar in a coil carrying alternating current (a.c.)
- D a steel bar in a coil carrying direct current (d.c.)

26. M/J/13/12/Q29

The magnetic field around two bar magnets is shown.

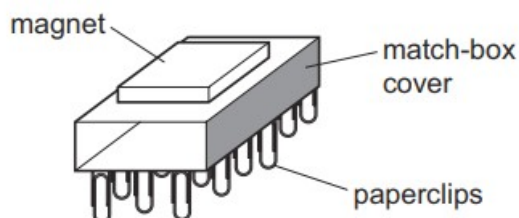


Which diagram represents the correct arrangement of magnetic poles?

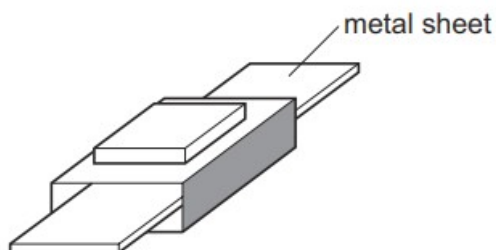
- | | | | | | | |
|---|---|---|---|---|---|---|
| A | <table border="1"><tr><td>N</td><td>S</td></tr></table> | N | S | <table border="1"><tr><td>N</td><td>S</td></tr></table> | N | S |
| N | S | | | | | |
| N | S | | | | | |
| B | <table border="1"><tr><td>N</td><td>S</td></tr></table> | N | S | <table border="1"><tr><td>S</td><td>N</td></tr></table> | S | N |
| N | S | | | | | |
| S | N | | | | | |
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| S | N | | | | | |
| S | N | | | | | |

27. M/J/13/11/Q26

A teacher sticks a magnet to the top surface of a match-box cover. The bottom surface is placed in a small tray of iron paperclips. As the match-box cover is lifted up, a large number of paperclips are held on the bottom surface.



Sheets of metal are placed inside the match-box cover, between the magnet and the paperclips.



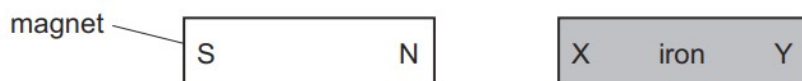
When sheet X is placed inside, the paperclips stay attached. When sheet Y is placed inside, all the paperclips fall off.

Which metals are the sheets made from?

	sheet X	sheet Y
A	aluminium	copper
B	copper	iron
C	iron	aluminium
D	iron	copper

28. O/N/12/12/Q26

A magnet is placed close to an iron bar.



The iron bar becomes an induced magnet.

Which magnetic poles are formed at the ends X and Y of the bar?

	end X	end Y
A	N	N
B	N	S
C	S	N
D	S	S

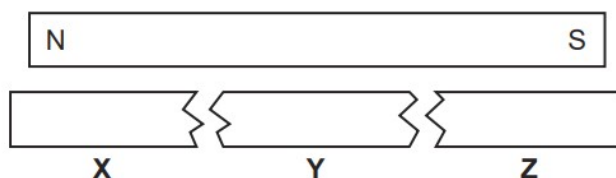
29. M/J/11/12/Q25

Which list contains an example of a non-magnetic material, a magnetic material and a magnetised material?

- A copper, iron, a compass needle
- B copper, iron, polythene
- C iron, steel, a compass needle
- D iron, steel, polythene

30. O/N/04/Q27

A bar magnet is broken into three parts **X**, **Y** and **Z**.



Which diagram shows the poles in **X**, **Y** and **Z**?

- A

X	Y	Z
N		S
- B

X	Y	Z
N	N	S S
- C

X	Y	Z
N	N	S N S S
- D

X	Y	Z
N	S	N S N S

ANSWERS

1. C
2. D
3. D
4. D
5. A
6. C
7. A
8. C
9. B
10. A
11. C
12. D
13. B
14. B
15. B
16. C
17. B
18. B
19. C
20. B
21. D
22. C
23. B
24. A
25. D
26. C
27. B
28. C
29. A
30. D

1. O/N/18/21/Q6

Fig. 6.1 is the view from above of a stationary magnet and a small compass on a laboratory bench.

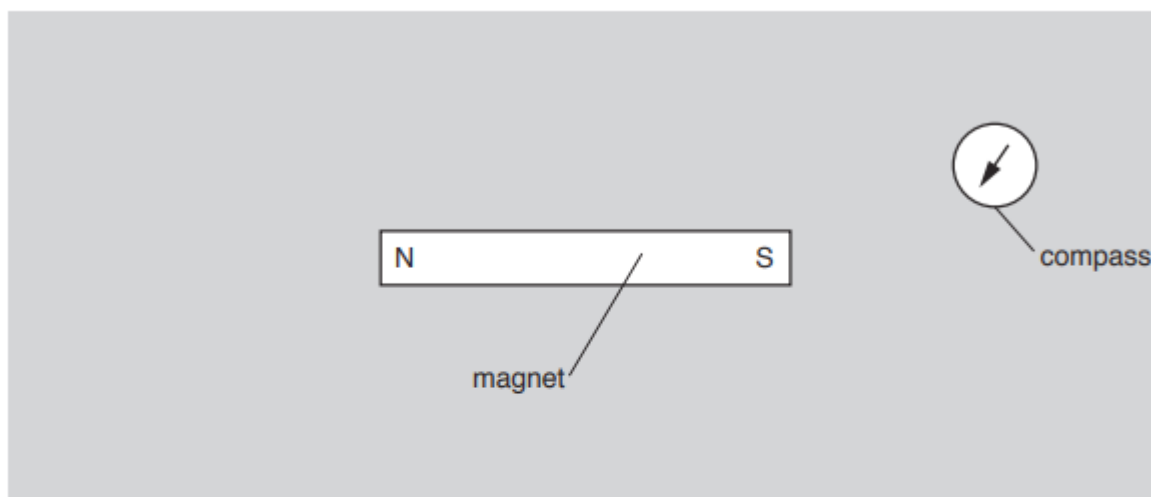


Fig. 6.1

- (a) State the material from which the compass needle is made.

..... [1]

- (b) Describe how the small compass is used to plot magnetic field lines in the region surrounding the magnet.

.....
.....
.....
.....
..... [3]

- (c) An unmagnetised iron bar PQ is placed near to the magnet on the laboratory bench.

Fig. 6.2 shows the two poles of the magnet and PQ.

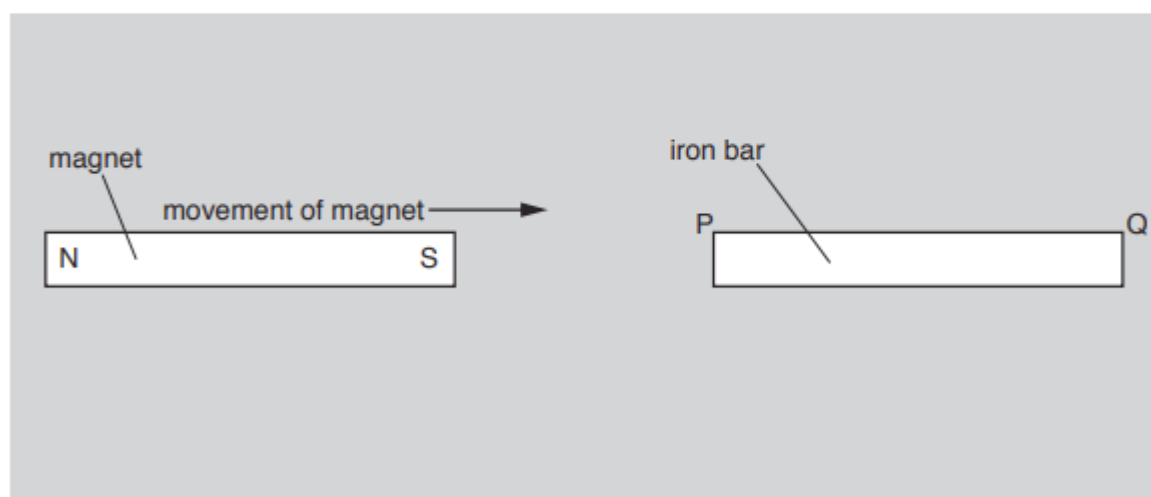


Fig. 6.2
15

The iron bar is initially at rest and the magnet is moved to the right very slowly.

When the magnet is a short distance from the iron bar, the iron bar moves very quickly to the left towards it.

Explain why the iron bar is attracted to the magnet.

.....

.....

.....

..... [3]

2. M/J/14/22/Q7

Fig. 7.1 shows a compass needle near a bar magnet. Magnetic poles are shown on the compass needle and on the magnet.

A finger stops the compass needle from turning.

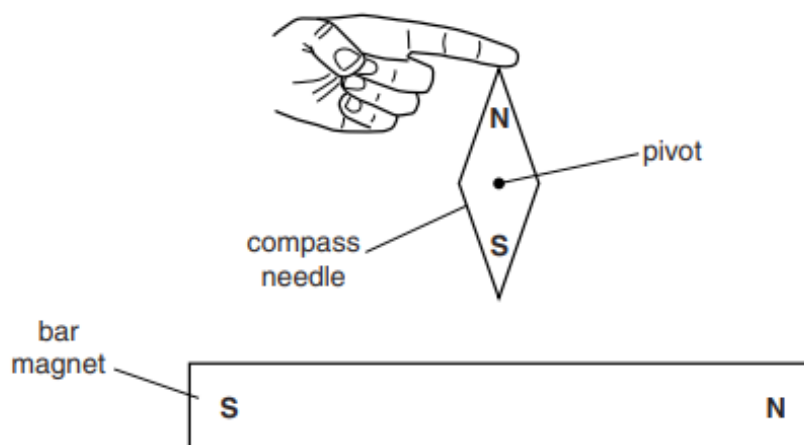


Fig. 7.1 (not to scale)

- (a) (i) The magnet causes a force on the S-pole of the compass needle.

On Fig. 7.1, draw an arrow from the S-pole of the compass needle to show the direction of this force. [1]

- (ii) Explain why the compass needle turns when the finger is removed.

.....
..... [1]

- (b) A small compass is used to plot the magnetic field lines of the magnet.

Describe how the compass is used to plot magnetic field lines on a piece of paper.

.....
.....
.....
.....
..... [3]

3. O/N/13/21/Q6

- (a) Suggest a material that is used to make a permanent magnet.

..... [1]

- (b) When there are no other magnetic fields present, the needle of a plotting compass points due north in the Earth's magnetic field. This is shown in Fig. 6.1.

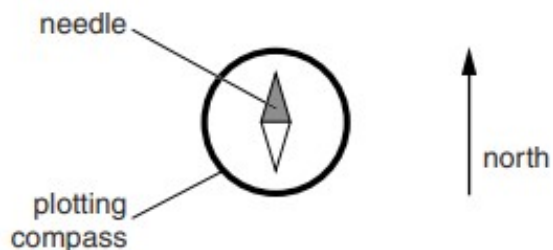


Fig. 6.1

The plotting compass is placed close to a permanent magnet, first at position A and then at position B, as shown in Fig. 6.2.

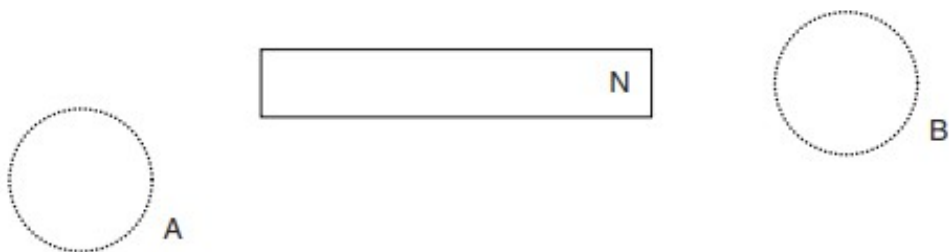


Fig. 6.2

The magnetic field close to the permanent magnet is very much larger than the magnetic field of the Earth.

On Fig. 6.2, draw needles in the two circles to show the direction in which the compass needle points when the compass is at A and at B. [2]

ANSWERS

1.

Question	Answer	Marks
6(a)	(hard) steel	B1
6(b)	(place compass near to a pole and) mark a dot at further end of needle	B1
	move compass so the first end coincides with the dot and mark second dot	B1
	repeat (many times) and join up dots	B1
6(c)	magnetism <u>induced</u> in iron bar or poles <u>induced</u> on iron bar	B1
	end P becomes an N-pole	B1
	unlike poles attract or (force of) attraction greater than friction / repulsion	B1

2.

(a) (i) **horizontal** arrow to right (by eye) [B1]

(ii) forces / resultant causes moment or (turns because) force is not at pivot [B1]

(b) mark made at one end/pole/direction of compass (on paper) [B1]

move compass so that other end of compass is on mark **and** remark [B1]

join marks made as compass moved on in some way (to draw line) [B1]

3.

(a) steel/alnico/SmCo/NdFeB/magnetite B1

(b) one needle fully correct **or** both angles correct – i.e. A bottom left to top right diagonal
($0 < \text{angle} < 90^\circ$) **and** B horizontal C1

both needles fully correct (fully = angle and orientation) A1

(c) (place) magnet in solenoid B1

a.c. supply to solenoid/coil (ignore cell/battery symbol) B1

withdraw magnet (slowly) **or** reduce current (slowly) B1